

Arif Cheng

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SKILLS

Languages

English, Mandarin
Chinese

Tech

Python, Java, Go,
PostgreSQL, Redis,
Apache Kafka, Docker,
Kubernetes, gRPC,
GraphQL

Tools

Git, AWS (EC2, RDS,
Lambda, SQS), GCP
(Cloud Run, Firestore),
Datadog, PagerDuty,
Terraform

ACTIVITIES

Berkeley Open

Source Club

Core Maintainer

2019 - 2020

Stripe Engineering

Conference

Speaker

Nov 2023

EDUCATION

University of California, Berkeley — *B.S. Computer Science*

2017 - 2020

- Completed coursework in Distributed Systems, Database Systems, and Systems Performance Engineering; GPA 3.7/4.0 with honors in Computer Science.

EXPERIENCE

Stripe — *Backend Engineer II*

Mar 2023 - Present

- Architected and deployed a distributed transaction reconciliation service handling 2.5M+ daily payment records, reducing reconciliation time from 8 hours to 12 minutes using event-driven Kafka pipelines.
- Optimized PostgreSQL query performance for the payments ledger by implementing materialized views and partitioning, decreasing p99 latency from 850ms to 120ms and supporting 40% increased throughput.
- Led migration of legacy XML-based API endpoint to gRPC-based microservice, reducing payload size by 65% and improving p95 latency by 200ms for 180M+ monthly requests across payment processing systems.

DoorDash — *Software Engineer, Backend*

Jul 2021 - Feb 2023

- Developed and maintained order state machine service processing 500K+ orders daily, implementing Redis-backed distributed caching that reduced database queries by 78% and improved API response time to sub-100ms.
- Implemented comprehensive monitoring and alerting system using Datadog and PagerDuty for 12 critical microservices, achieving 99.94% uptime SLA and reducing MTTR by 35%.
- Designed async job processing system using Apache Kafka and worker pools for restaurant onboarding, scaling from 5K to 50K+ restaurants and reducing processing latency from 2 hours to 8 minutes.

Figma — *Backend Software Engineer*

Jan 2021 - Jun 2021

- Built real-time collaboration backend using WebSocket and Operational Transformation principles, supporting 100K+ concurrent editing sessions with sub-50ms sync latency.
- Refactored database schema for collaborative document storage, eliminating N+1 query patterns and reducing storage footprint by 42% while maintaining backward compatibility across 2M+ documents.
- Implemented rate limiting and quota enforcement middleware in Go, protecting critical APIs from abuse and reducing infrastructure costs by \$180K annually through better resource utilization.

PROJECTS

High-Throughput Event Processing Pipeline — *Go, Apache*

Apr 2023 - Jun 2023

Kafka, PostgreSQL, Prometheus

- Engineered fault-tolerant event aggregation system processing 1M+ events/second with exactly-once semantics, utilizing consumer groups and offset management to maintain data consistency under network partitions.
- Implemented distributed tracing using OpenTelemetry to identify bottlenecks, achieving 40% reduction in end-to-end latency and enabling real-time visibility into processing pipeline health.

Multi-Region Database Failover Orchestrator — *Python,*

Sep 2022 - Dec 2022

Terraform, AWS RDS, etcd

- Developed automated failover system for PostgreSQL clusters across 3 AWS regions, reducing RTO from 45 minutes to 3 minutes and enabling zero-downtime migrations for 12 critical services.
- Created comprehensive health-check framework with consensus-based decision making, preventing split-brain scenarios and ensuring data consistency across 500GB+ databases.

GraphQL API Gateway — *GraphQL, Java, Redis, Docker*

Feb 2021 - May 2021

- Developed unified GraphQL gateway aggregating 8 downstream microservices, reducing client-side API calls by 72% and enabling declarative data fetching for 15 mobile and web clients.

- Implemented intelligent request batching and caching layer using Redis, decreasing median query resolution time from 580ms to 140ms and reducing backend service load by 55%.